

Next, we look at the differential forms. The general form of the equations is

$$\begin{aligned}\frac{\partial(\rho u)}{\partial t} + \nabla \cdot (\rho u \vec{V}) &= -\frac{\partial p}{\partial x} + \rho f_x + (f_x)_\mu \\ \frac{\partial(\rho v)}{\partial t} + \nabla \cdot (\rho v \vec{V}) &= -\frac{\partial p}{\partial y} + \rho f_y + (f_y)_\mu \\ \frac{\partial(\rho w)}{\partial t} + \nabla \cdot (\rho w \vec{V}) &= -\frac{\partial p}{\partial z} + \rho f_z + (f_z)_\mu\end{aligned}$$

(a) For a steady flow then $\partial/\partial t \equiv 0$. So, the reduced form is

$$\begin{aligned}\nabla \cdot (\rho u \vec{V}) &= -\frac{\partial p}{\partial x} + \rho f_x + (f_x)_\mu \\ \nabla \cdot (\rho v \vec{V}) &= -\frac{\partial p}{\partial y} + \rho f_y + (f_y)_\mu \\ \nabla \cdot (\rho w \vec{V}) &= -\frac{\partial p}{\partial z} + \rho f_z + (f_z)_\mu\end{aligned}$$

(b) For a steady, compressible, inviscid flow then $\vec{F}_\mu = 0$ so:

$$\begin{aligned}\nabla \cdot (\rho u \vec{V}) &= -\frac{\partial p}{\partial x} + \rho f_x \\ \nabla \cdot (\rho v \vec{V}) &= -\frac{\partial p}{\partial y} + \rho f_y \\ \nabla \cdot (\rho w \vec{V}) &= -\frac{\partial p}{\partial z} + \rho f_z\end{aligned}$$

(c) For a steady, incompressible, viscous flow then

$$\begin{aligned}\rho \nabla \cdot (\vec{V}) &= \vec{V} \cdot \nabla \rho \\ \rho \nabla \cdot (\vec{V}) &= -\frac{\partial p}{\partial x} + \rho f_x + (f_x)_\mu \\ \rho \nabla \cdot (\vec{V}) &= -\frac{\partial p}{\partial y} + \rho f_y + (f_y)_\mu \\ \rho \nabla \cdot (\vec{V}) &= -\frac{\partial p}{\partial z} + \rho f_z + (f_z)_\mu\end{aligned}$$

(d) For a steady, incompressible, inviscid, two-dimensional flow then

$$\begin{aligned}\vec{V} \cdot \nabla u &= u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + f_x \\ u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} &= -\frac{1}{\rho} \frac{\partial p}{\partial y} + f_y\end{aligned}$$

(e) For a compressible, inviscid, one-dimensional flow then

$$u \frac{d(\rho u)}{dx} = -\frac{dp}{dx} + f_x$$

$$\vec{V} \cdot \nabla u = u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = u u_x + v u_y$$

Continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{V}) = 0$$

$\rho = \text{cte}$

$$\Rightarrow \nabla \cdot \vec{V} = 0$$

From continuity

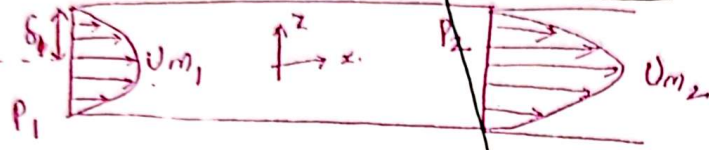
$$\nabla \cdot \vec{V} = 0$$

$$\nabla \cdot (u \vec{V}) = u \nabla \cdot \vec{V} + \vec{V} \cdot \nabla u = 0$$

math rule

(unsteady term) $+ \iint_S \rho \vec{V} \cdot d\vec{S} = - \iint_S p dS + D_w \Rightarrow D_w = \iint_S p dS$

$$D_w = A(p_1 - p_2)$$



Continuity Equation:

$$\frac{\partial}{\partial t} \iiint_V \rho dV + \iint_S \rho \vec{V} \cdot d\vec{S} = 0$$

Assumptions:

Steady, incompressible flow: No body forces.

$$\Rightarrow \iint_S \vec{V} \cdot d\vec{S} = 0$$

$$-\iint_{S_1} u_{m1} dS_1 + \iint_{S_2} u_{m2} dS_2 = 0$$

$$-u_{m1} S_1 + u_{m2} S_2 = 0 \quad [A_1 S_1 = A_2 S_2]$$

$$\Rightarrow \boxed{u_{m1} = u_{m2}} \quad \text{--- (1)}$$

Now, Momentum Equation is given by.

$$\frac{\partial}{\partial t} \iiint_V \rho \vec{V} dV + \iint_S (\rho \vec{V} \cdot d\vec{S}) \vec{V} = \iiint_V \rho \vec{f} dV - \iint_S p d\vec{S} + F_{ext}$$

Now as $u_{m1} = u_{m2}$

$$\Rightarrow \iint_S \rho (\vec{V} \cdot d\vec{S}) \vec{V} = 0$$

Net mom. is zero
exchange

$$\Rightarrow F_{ext} = \iint_S (\vec{p} \cdot \hat{n}) dS$$

$$\Rightarrow F_{fl} = -F_{ext} = -\iint_S (\vec{p} \cdot \hat{n}) dS = -(p_1 A_1 + p_2 A_2)$$

$$\Rightarrow \boxed{F_{fl} = A(p_1 - p_2)} \quad \text{--- (2)}$$

$$\iint_{m_1} u dS = \iint_{m_2} u dS$$

$$\boxed{u_{m1} = u_{m2}}$$

$$\overline{u_{m1}} = \overline{u_{m2}}$$

$$\underbrace{\int u_{m1} u_{m1}}_{\text{mom}_{in}} = \underbrace{\int u_{m2} u_{m2}}_{\text{mom}_{out}}$$